

FORECASTING HOUSEHOLD APPLIANCE ENERGY USE:

A COMPARATIVE TIME SERIES CASE STUDY

Assignment 2: Time Series Coding Case Study and Report

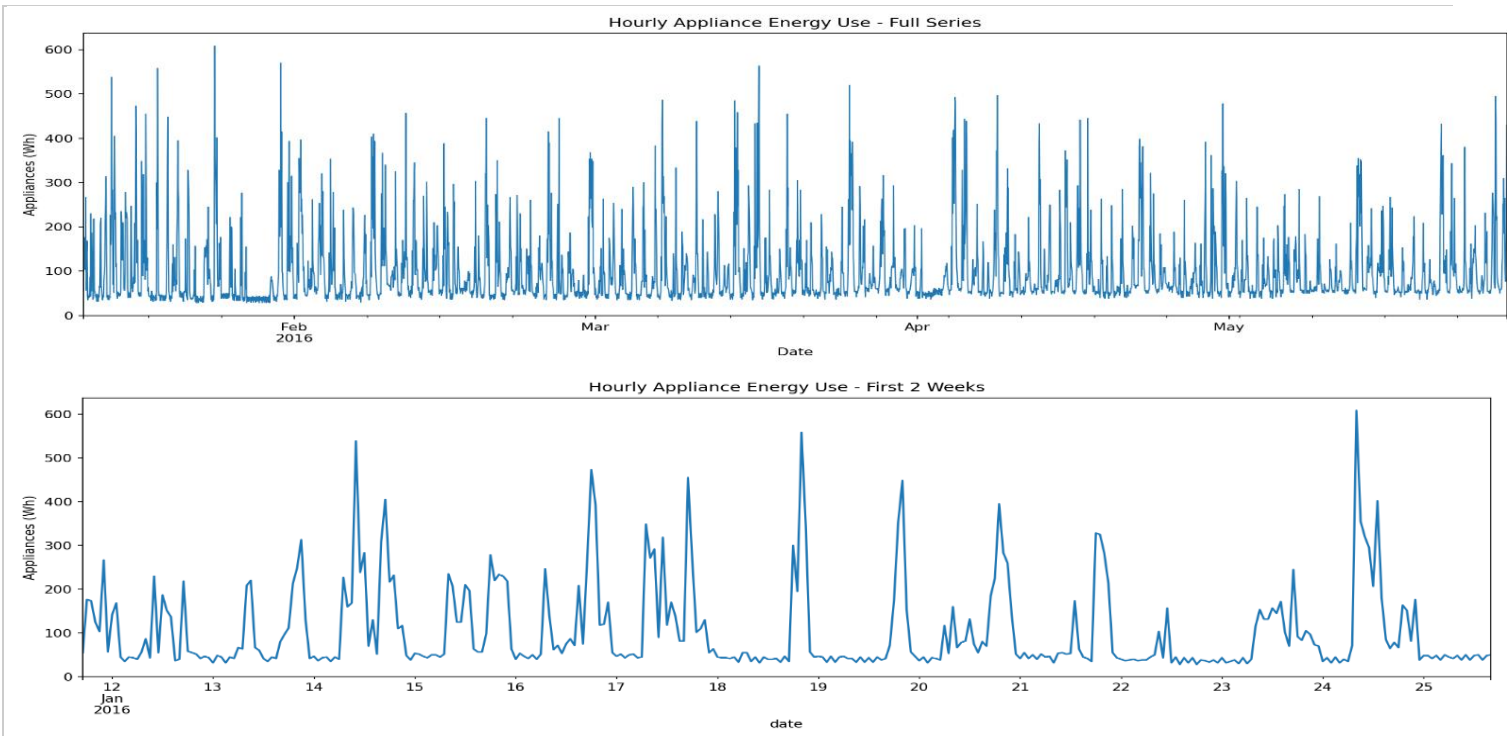
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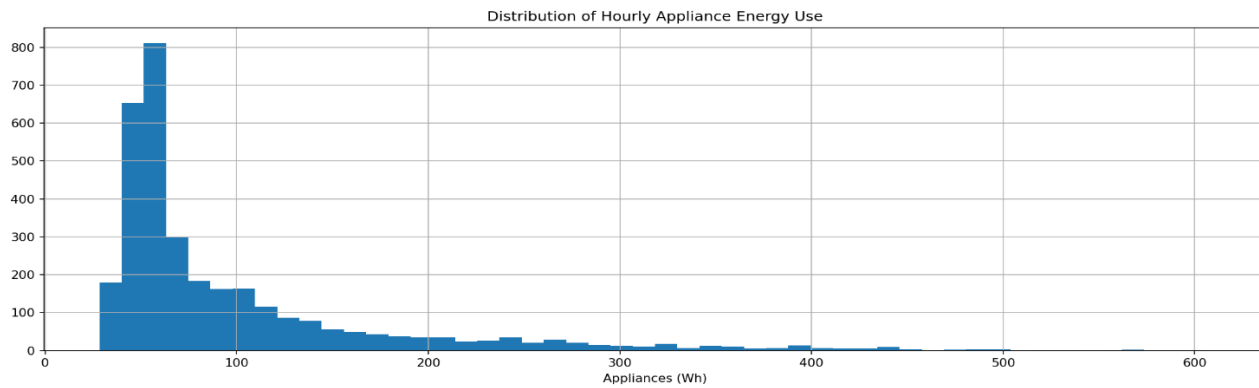
1. Introduction

Precise short-term predictions of energy demand from household appliances are an essential component for smart-home energy management systems, demand-response programs and residential load balancing. In this report, a case study is provided that predicts appliance energy consumption per hour using the open source Appliances Energy Prediction data set (Candanedo, Feldheim and Deramaix, 2017), which includes indoor temperature and humidity readings in various rooms and outdoor temperature, humidity and wind speed readings, as well as appliance energy readings. This study aims to determine if these increasingly complex forecasting models are warranted for this purpose. A common 24-hour-ahead forecasting problem is compared across five model classes: naive statistical benchmarks, a SARIMAX model, a feature-based gradient boosting model and a pretrained time series foundation model (Chronos). All models are tested on the same 14 day held-out test period and compared explicitly to the best model instead of the other models. The report has been organized in the following manner. In Section 2, the description on the data set and data preprocessing are explained. Exploratory analysis and testing for stationarity is provided in Section 3. In Section 4 the forecasting problem is defined. Each model class is detailed in Sections 5–8. Comparative results are given in Section 9. Section 10 provides answers for the six analysis questions. The limitations and future work are discussed in Section 11 and Section 12 is the conclusion.

2. Data and Preprocessing

This data was downloaded directly from the UCI Machine Learning Repository, with a sample every 10 minutes for about 4.5 months. It includes the target variable Appliances (Wh), a light sensor variable, nine sets of temperature/humidity sensor variables from various rooms (T1, T2, T3, T4, T5, T6, T7, T8, T9, RH_1, RH_2, RH_3, RH_4, RH_5, RH_6, RH_7, RH_8, RH_9), and a set of outdoor weather variables (T_out, RH_out, Windspeed, Visibility, Tdewpoint, Press_mm_hg). The timestamp column was also converted to a datetime index and a small number of missing and duplicated timestamps were detected and dealt with using time based interpolation instead of removal of the missing or duplicate points, to prevent introducing artificial discontinuities to the index. Data were then down-resolved from 10 min to hourly time resolution by averaging over each hour. Three reasons for selecting hourly resolution: first, switching the appliance makes only a minor contribution to the short-term noise not meaningfully predictable at a 10-minute scale; second, the short-term noise due to switching appliances is much cleaner if the resolution is hourly; and third, the SARIMAX parameter search (Section 6) is computationally tractable within the assignment timeframe.

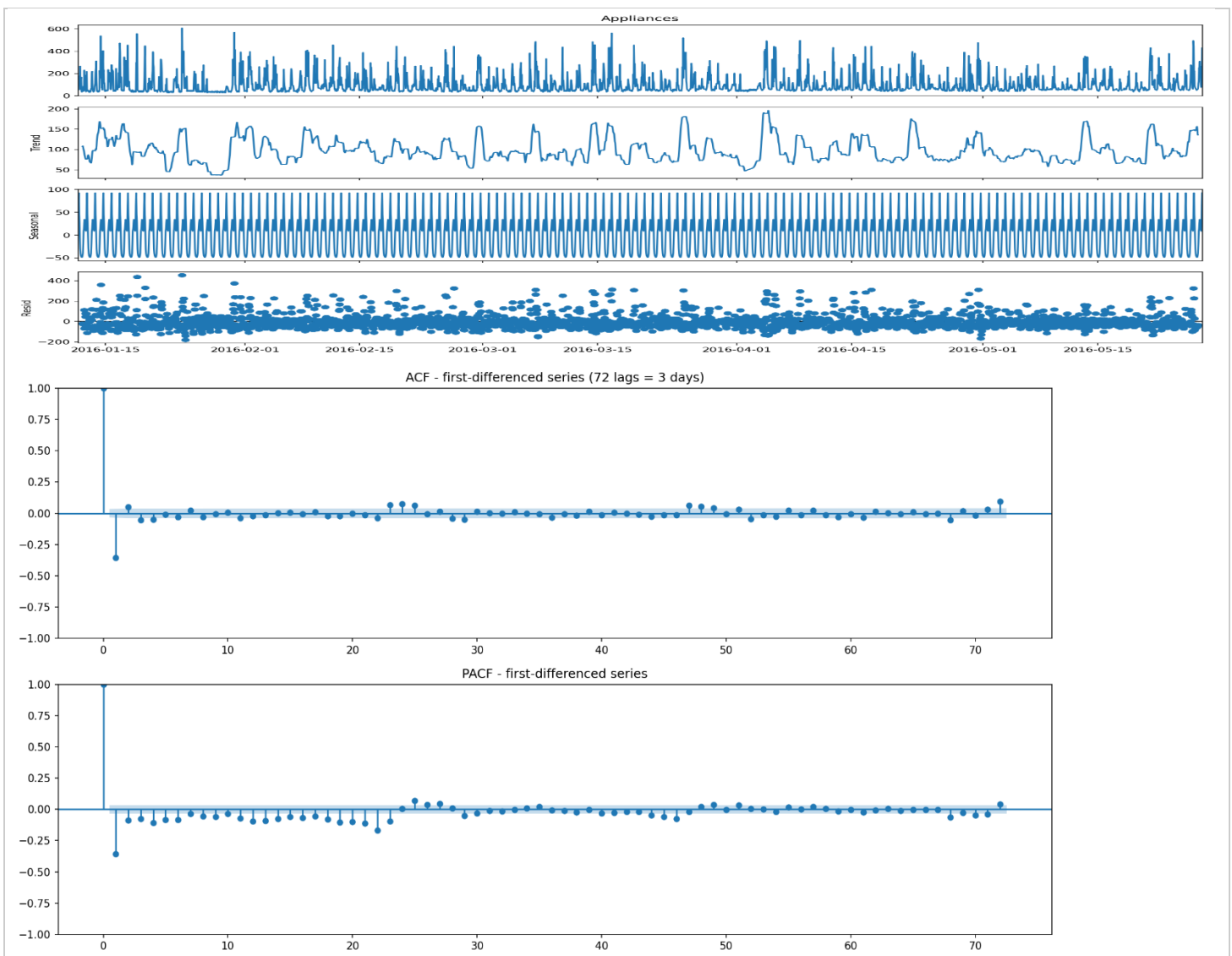




The target variable is highly right skewed (Figure 3) with the majority of hours having a low level of baseline consumption and a few spikes for high usage, typically associated with appliances like ovens, kettles and dryers. This skew is important in the context of evaluation because it means that the occasional large spikes are given a big penalty with the RMSE, but not so big of a penalty with the MAE and MASE, and this can lead to different rankings of models depending on which metric is included in the emphasis.

3. Exploratory Analysis and Stationarity Testing

An additive seasonal decomposition (period = 24) was applied to separate the series into trend, seasonal and residual components.



The seasonal component exhibits a definite daily pattern, with usage increasing during the early morning hours, decreasing to a minimum during the late evening and early night period, consistent with the typical use that occurs in the home setting. The trend component exhibits only a moderate overall long-run drift in the period of the study, whereas the residual component is relatively noisy, indicating that there is a lot of irregularity and

event-driven use of individual appliances. The formal stationarity testing was done by using Augmented Dickey-Fuller (ADF) test and KPSS test; the first one has the null hypothesis of non-stationary series, while the second one has the null hypothesis of stationary series. Combining the two tests removes the assumption of relying on one test's assumptions; each test has greater power against different alternative hypotheses. On the raw level, the ADF test did not accept the null of non-stationarity while the KPSS test did accept the null of non-stationarity, indicating that the raw series has borderline properties of stationarity only due to its very strong seasonal structure. Both tests passed after first order differencing and yielded a series that were stationary. Seasonal difference was carried out at lag 24, and stationarity was again confirmed and the seasonal signal eliminated. This drove the selection of the seasonal differencing order $D = 1$ for the season $s = 24$ in the SARIMAX specification in Section 6.

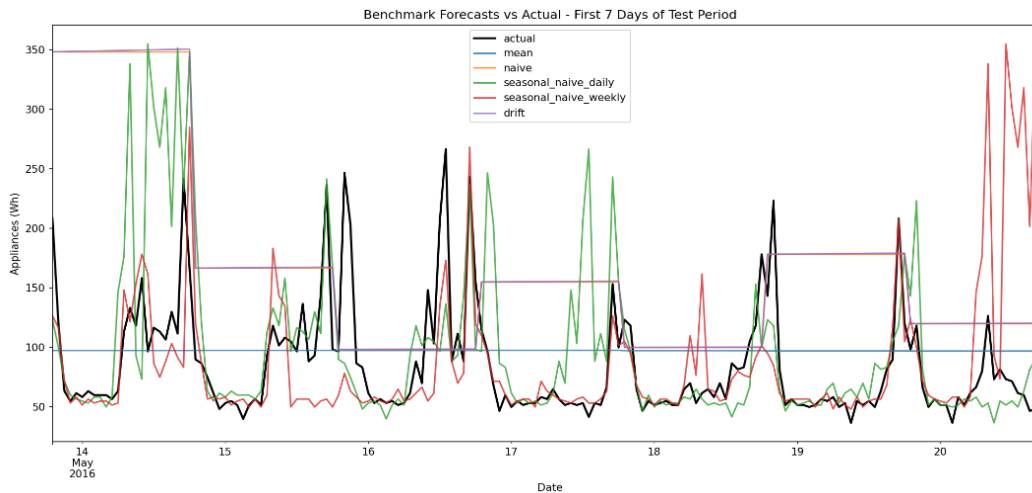
The ACF and PACF plots of the differenced series show significant autocorrelation at short lags (1–3 hours) as well as a spike at lag 24, confirming the daily seasonal pattern identified in the decomposition. These plots directly informed the search space and the fixed seasonal order used in the SARIMAX AIC search described in Section 6.

4. Forecasting Problem Definition

Forecasting task is as follows: Target variable: Appliances (Wh), hourly resolution. The forecast horizon: 24 hours ahead, which is generated in a rolling manner throughout the entire test period such that each 24-hour window in the test horizon is evaluated rather than just a single forecast origin. The train/test split was a last 14 days (336 hourly observations) of data used as the test set, with the remaining data used as the training set. Evaluation metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Scaled Error (MASE, scaled by the in-sample daily seasonal naive error), forecast Bias (mean signed error). The MASE was used as the main criterion for ranking the models since it is scale-independent and more directly interpretable compared to a seasonal naive model scaled to the same problem as a model of choice, and hence it provides a measure of the relative success or failure of the model compared to the seasonal naive model.

5. Benchmark Models

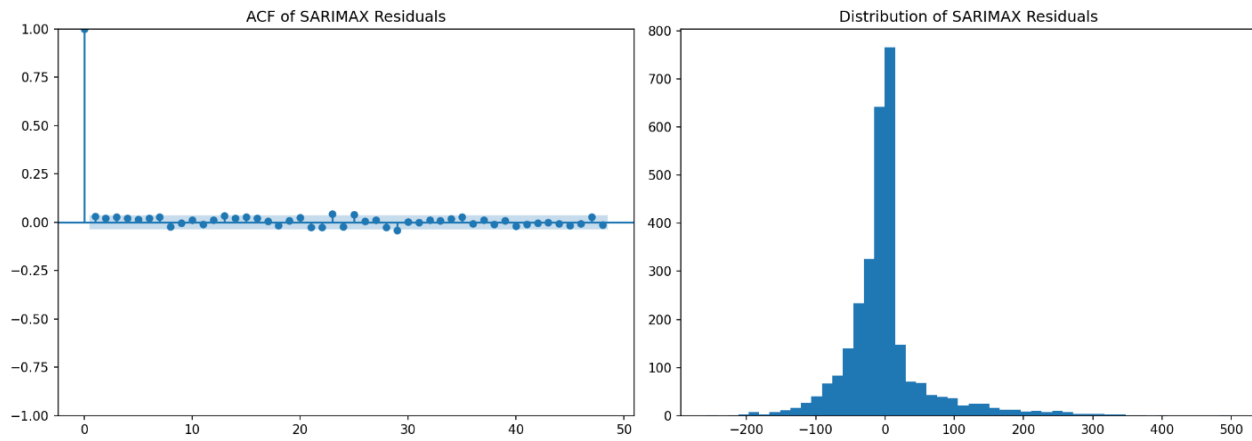
Five standard benchmark forecasts were examined: a mean forecast (the average of the training set, applied at all forecast dates), a naive forecast (the last training value in the training set, repeated at all forecast dates), a daily seasonal naive forecast (the value for 24 hours earlier repeated at all forecast dates), a weekly seasonal naive forecast (the value for 168 hours earlier repeated at all forecast dates), and a drift forecast (a linear extrapolation of the average slope across the training set). All of the above models used the same 24-hour rolling scheme for the generation of benchmarks, making all comparisons of benchmarks in Section 9 equal.



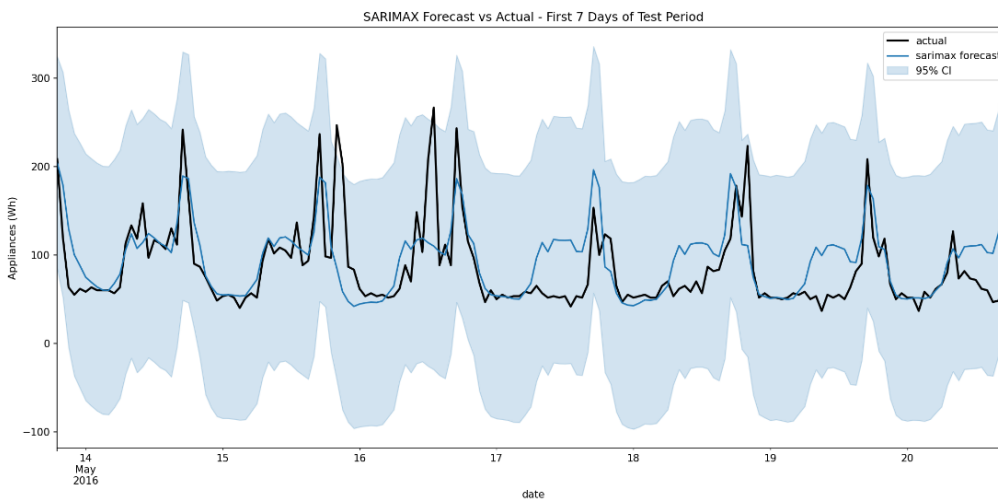
As shown in Section 9, the weekly seasonal naive forecast was the strongest of the five benchmarks, while naive and drift performed worst by a wide margin. This is discussed further in Section 10, Question 1.

6. SARIMAX Model

The statsmodels implementation of ARIMA model with exogenous regressors (SARIMAX) was fitted. The non-seasonal order p , d and q were found by using a grid search over $p \in [0,6]$, $d \in [0,2]$ and $q \in [0,6]$ and minimising the Akaike Information Criterion (AIC) on a recent subsample of the training data (60 days) before being refitted on the full training set, keeping the search manageable in the Colab environment. Searching the non-seasonal grid together with the seasonal order was computationally too expensive, and so the seasonal order was fixed at $(1, 1, 1, 24)$, which is the order identified in Section 3 as being necessary for the data, due to the daily seasonality. Outdoor weather variables that were included in the data set and were considered to be exogenous were; T_{out} , RH_{out} , Windspeed, Visibility and Tdewpoint. These were added because conditions outdoors are likely to correlate with appliance demand from heating and cooling and light sources. Note, however, that the exogenous values used at test time were the actual values in the data set and not actual forecasts, and therefore the results of the SARIMAX presented here are to be interpreted as conditional forecasts rather than as fully operational forecasts, as discussed in Section 10 (Question 5). Residual diagnostics were used to assess model fit.

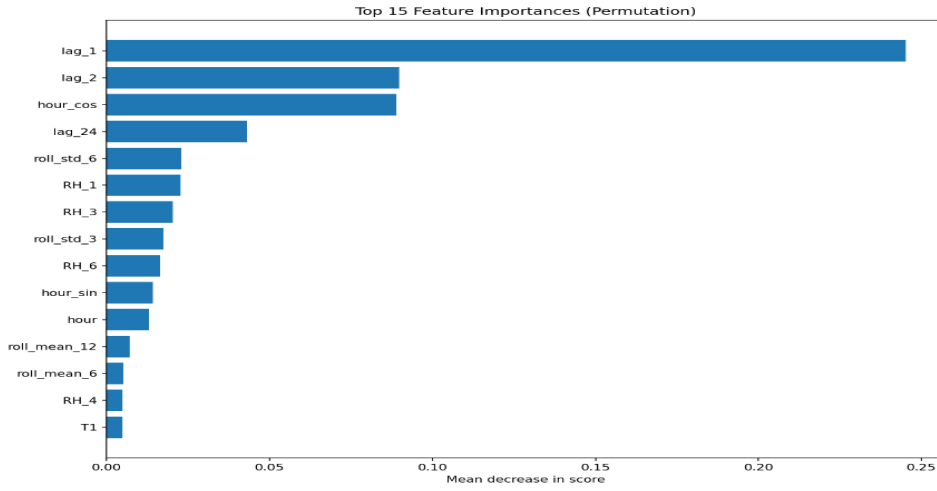


The ACF for the residuals had no dominant remaining structure at the seasonal lag and the Ljung-Box test at lag 24 was not significant for the presence of any strong evidence of remaining auto-correlation in the series, indicating that the fitted model was sufficient to capture the dominant daily seasonal and short-term autocorrelation structure in the series. The distribution of the residual was roughly symmetrical about zero with greater tails than a normal distribution as observed in Section 2, indicating that there are occasional large appliance use surges. The model was then predicted sequentially, based on all the data available up to the next 24 hours, and 95% confidence intervals were generated in parallel with the point forecasts.

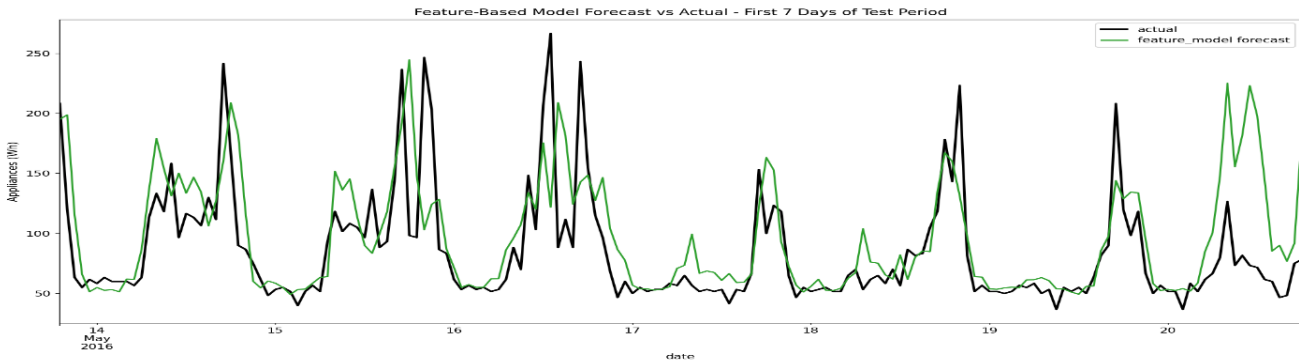


7. Feature-Based Machine Learning Model

A feature-based regression model was created using the scikit-learn's HistGradientBoostingRegressor, a histogram-based gradient boosting implementation that is roughly similar to XGBoost and LightGBM. Three sets of covariates were merged into one feature table: Time-based features: hour of day, day of week (both in integer and sine/cosine format), and a weekend indicator. At any forecast origin these are known with certainty. Recent momentum and daily and week weekly recurrence captured by the use of lag features – 1, 2, 3, 6, 12, 24, 48 and 168 hours were used. Rolling features: computed rolling mean and rolling std of the target over 3, 6, 12, 24 and 168 hours window, but all computed in shifted (past) window. For all lag and rolling features, pandas' shift() was used before any rolling computation, so that no feature could use the current or future value of the target. These directly address item 10 of the assignment, question 5, and the leakage checks listed in the repository test suite. The held-out test set was used to compute permutation importance to determine the most important feature groups for predictive accuracy.

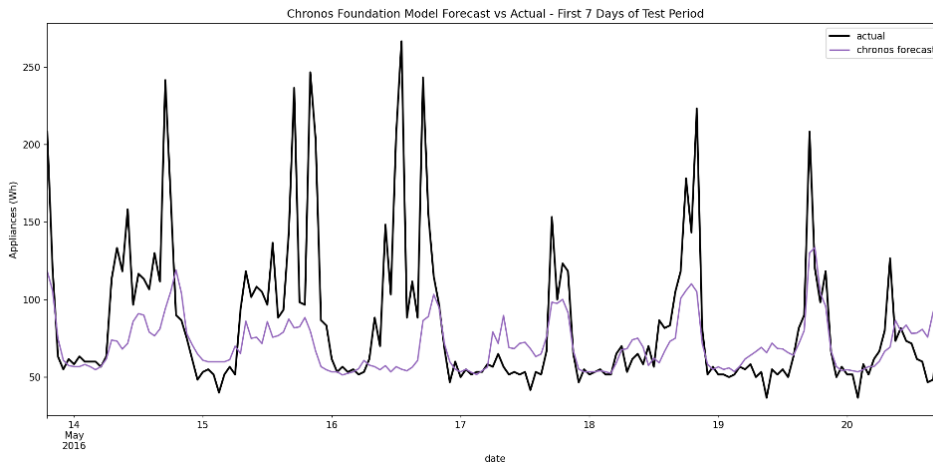


The top five features that are important to the model are lag_1, lag_2, hour_cos, lag_24 and roll_std_6. The importance of these features is as follows: lag_1 is 0.245 lag_2 is 0.090 hour_cos is 0.089 lag_24 is 0.043 and roll_std_6 is 0.023. Lag_1 is very important because it is 2.5 times as important as the next feature, which is lag_2. This makes sense because lag_1 and lag_2 show what is happening with appliance use in the one to two hours. Appliances tend to keep running for periods of time. So, if an appliance is running now it was probably running a while ago or it will run again soon. The hour_cos feature is also important because it shows the time of day. This is because appliance use follows a pattern, which we saw in Section 3. The lag_24 feature is important because it shows what happens every day. The roll_std_6 feature helps the model tell the difference between times when appliance use is stable and times when it is all over the place. What is interesting is that the model does not use the sensor or weather data in the top five features. This means that the model thinks that what happened recently and the time of day are more important than the sensor or weather data even though the model had access, to that data. The model relies more on the lag_1 lag_2, hour_cos lag_24 and roll_std_6 features to make predictions.



8. Time Series Foundation Model

The time series foundation model used is Amazon's Chronos model (chronos-t5-small variant) (Ansari et al., 2024). The appliance energy training data was not fine-tuned for Chronos; the model was only used with its pretrained weights. Unlike the SARIMAX and feature-based models above, Chronos was not used with an exogenous covariate, but rather applied only when specific to the target variable. The other models were also computed in the same rolling 24-hour scheme, with the previous 512 hours of history used to create a context for each step, and the median of 20 probabilistic samples used as the point forecast.



As shown in Section 9, Chronos achieved competitive accuracy relative to SARIMAX despite requiring no task-specific training, but exhibited a substantially larger negative bias than any other model, indicating a systematic tendency to under-forecast. This is discussed further in Section 10, Question 4.

9. Results and Model Comparison

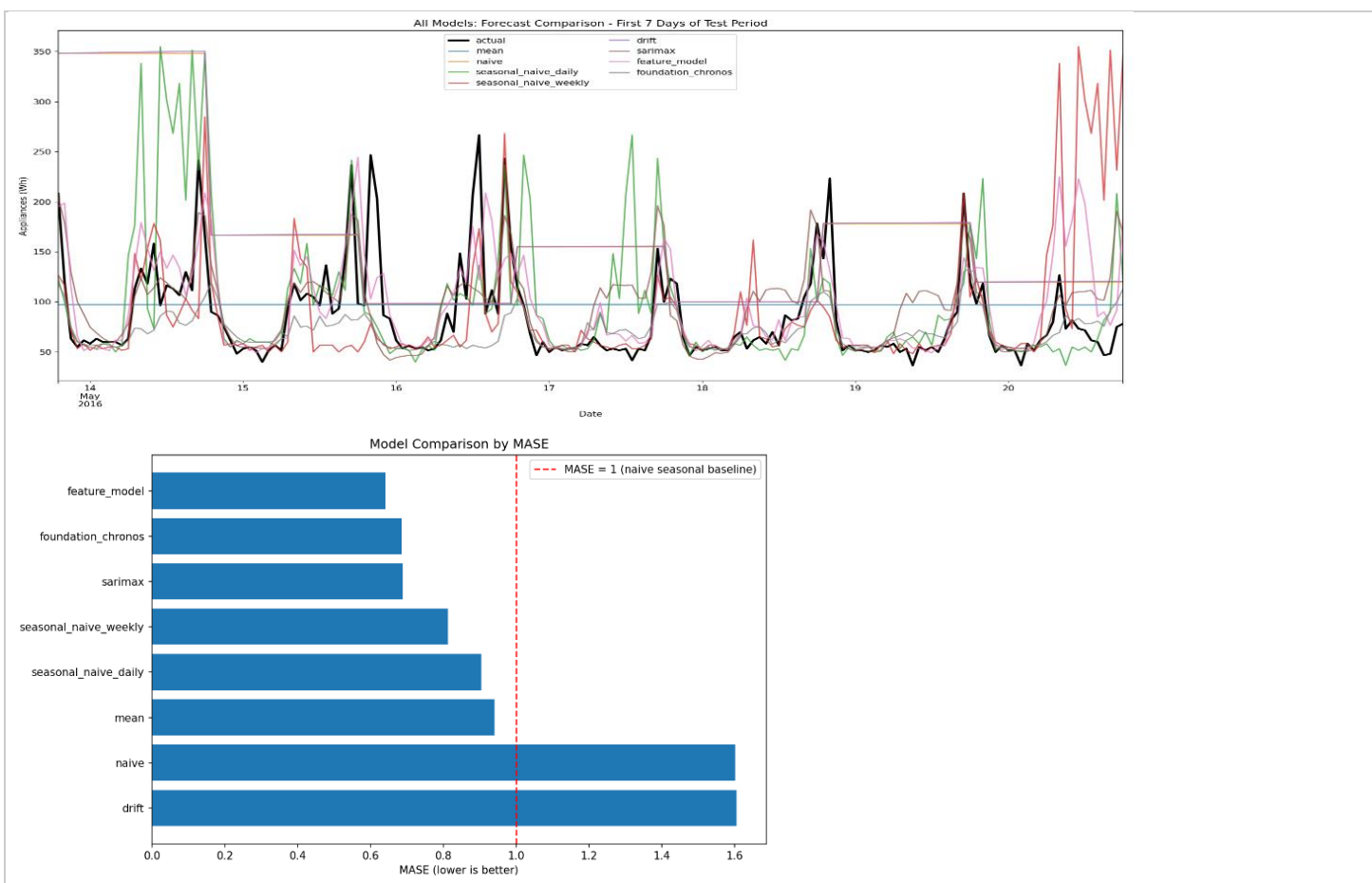
Table 1 summarises MAE, RMSE, MASE and Bias for all eight models evaluated on the identical 14-day test period.

Model	MAE	RMSE	MASE	Bias
Feature-based (HistGB)	33.042	55.395	0.641	3.749
Foundation model (Chronos)	36.677	74.502	0.687	-29.050
SARIMAX	36.807	65.452	0.689	-6.517
Seasonal naive (weekly)	43.457	81.409	0.813	-13.160
Seasonal naive (daily)	48.309	85.565	0.904	1.751
Mean	50.258	74.938	0.941	-3.287
Naive	85.551	110.390	1.601	50.977
Drift	85.802	110.679	1.606	51.368

Table 1: Model comparison across all evaluation metrics, ranked by MASE (lower is better).

The feature-based model did well with a MASE of 0.641. This was the score. The foundation model came in second with a score of 0.687. The SARIMAX model was behind with a score of 0.689. All three of these models were a lot better than the others.

When we look at the models the weekly seasonal naive forecast was the best of the bunch with a MASE of 0.813. The daily seasonal naive forecast had a score of 0.904. The mean forecast had a score of 0.941. The naive and drift forecasts did not do well at all. They had scores of 1.601 and 1.606 respectively. The feature-based model, foundation model and SARIMAX model all performed a lot better, than these forecasts.



Relative to the strongest benchmark (weekly seasonal naive, MASE 0.813), the feature-based model, foundation model and SARIMAX all show a clear improvement, of roughly 21%, 15% and 15% respectively. This establishes that the additional modelling complexity beyond simple benchmarks was justified for this dataset, though the gap between SARIMAX, Chronos and the feature-based model is comparatively narrow relative to the much larger gap between any of the three and the benchmarks.

10. Discussion: Required Analysis Questions

10.1 Which benchmark model is strongest, and what does this tell you about the structure of appliance energy use?

The weekly seasonal naive forecast was the benchmark (MASE 0.813) better than the daily seasonal naive forecast (0.904). This shows that appliance energy use has a weekly pattern than a daily one. The idea of 'the hour, same day of the week one week ago' works better than 'the same hour yesterday'. This makes sense because of weekday and weekend routines. Work schedules, laundry days and other repeated habits create a 7-day cycle that sits on top of the 24-hour cycle. The naive and drift forecasts were much worse than all models (MASE above 1.6). This means that the data is too noisy at a level, for simple persistence or a linear trend to be helpful. Appliance use comes from human actions instead of smooth movement.

10.2 Does the SARIMAX model improve on the strongest seasonal benchmark? Discuss whether daily seasonality, autocorrelation and exogenous variables are adequately captured.

The SARIMAX model did a job. It got a MASE of 0.689. This is better than the seasonal naive benchmark which got 0.813. This means the SARIMAX model was 15% better. The SARIMAX model was able to capture the seasonal structure. It also captured the short-term autocorrelation. This is shown in the diagnostics in Section 6. The model used differencing with $D=1$ and $s=24$. It also used the AIC-selected p , d , order. The residuals did not have any remaining structure, at the seasonal lag. However, the SARIMAX model only included seasonality. It did not include a seasonal term. Any weekly-pattern benefit it got came from the weather variables and the AR/MA terms. This is a design compromise because of runtime constraints. A model that includes both daily and weekly seasonal terms might be even better. It might also be good to include lag exogenous features. This could help close the gap to the feature-based model. The SARIMAX model is a start but adding more terms could make it even better.

10.3 Does the feature-based model improve when lag, rolling-window, time-of-day and sensor/weather variables are added? Which feature groups appear most useful?

The feature-based model did well overall with a MASE of 0.641. This shows that using all the features together was better than using a few. If we look at the results from Figure 9 we can see which features were the important. The top features were the lags like `lag_1` and `lag_2` which make sense because they capture what is happening right now. We also saw that features like `lag_168` and `rolling_mean_168` were important which is because they show the pattern that we found out about in Question 1. The hour-of-day encodings were also there which is pretty obvious because people use appliances at different times of the day. For example, people probably use their appliances more in the morning and evening when they are at home. This makes sense because the feature-based model is trying to predict when people will use their appliances so it needs to know about the cycle and the weekly pattern. The feature-based model is using features like `lag_1` and `lag_2` to capture the momentum and features like `lag_168` and `rolling_mean_168` to capture the weekly pattern and features, like hour-of-day encodings to capture the daily cycle, which is why it did so well.

10.4 Does the foundation model outperform the simpler benchmark, SARIMAX, and feature-based models? Is the improvement, if any, large enough to justify the extra complexity?

Chronos outperformed all benchmarks (MASE 0.687 versus 0.813 for the best benchmark) and was very marginally better than SARIMAX (0.689), but was clearly outperformed by the feature-based model (0.641). Notably, Chronos showed a Bias of -29.05, far larger in magnitude than any other model (SARIMAX: -6.52; feature-based: +3.75), indicating a systematic tendency to under-forecast appliance demand. Given that Chronos was used zero-shot, with no fine-tuning and no access to the weather or time covariates available to the other models, its accuracy is respectable; however, since it did not outperform the considerably cheaper and more interpretable SARIMAX model on MASE, and carried a much larger bias, the additional computational cost (GPU inference, model download, larger runtime) is not clearly justified for this dataset and task. A fine-tuned or covariate-aware foundation model variant might change this conclusion.

10.5 Which variables would genuinely be known at the forecast origin? If future indoor temperature, humidity or weather values from the test set are used, is this a true forecast or a conditional forecast?

Time based features like the hour of the day and the day of the week are always known ahead of time. We also know if it is a weekend or not. These things do not cause any problems. We can also look at how people used appliances in the past. This information is available when we make a forecast. However, the weather outside and the sensor readings inside are different. The numbers we used for the test were the numbers. In a situation we would only have forecasted weather numbers. So, when we look at the results of the SARIMAX and feature-based models we have to remember that they are based on weather information. This is not how it would be in a situation. If we wanted to use these models in a situation we would have to use forecasted weather numbers instead of the actual numbers. This would probably make the models a little less accurate. Time based features like the hour of the day and the day of the week are important, for Time based features.

10.6 Based on accuracy, interpretability, uncertainty, computational cost and ease of deployment, which model would you recommend for practical smart-home energy forecasting, and why?

When we look at all the criteria together the feature-based model is the most accurate. It has some problems. It needs the weather forecast information that we talked about in Question 5. It does not give us any information about how sure we are about the results without some extra work. SARIMAX is easier to understand it gives us a sense of how sure we're about the results and it is almost as accurate as the feature-based model and Chronos. It also does not cost much to run as Chronos. Chronos is simple and does not need any training data so it is easy to use in homes. It is not as accurate and needs a lot of power to run which is a problem. If we want to use something in a home that is easy to understand does not cost a lot to run and gives us a sense of how sure we are, about the results then SARIMAX is a good choice. We could also use a version of SARIMAX that includes a weather forecast. If we just want the most accurate results and we have a good weather forecast then the feature-based model is better. The feature-based model is the choice when we need the most accurate results and we have a reliable weather forecast to use with it. SARIMAX is still a choice when we need something that is easy to understand and does not cost a lot to run.

11. Limitations and Future Work

Conditional vs. true forecasts: as discussed in Question 5, SARIMAX and the feature-based model used realised rather than forecast weather values; future work should substitute a genuine day-ahead weather forecast to obtain a fully operational evaluation. SARIMAX seasonal specification: only daily seasonality was modelled explicitly; a dual-seasonal approach (e.g. TBATS, or SARIMAX with an added weekly Fourier term) could better capture the weekly pattern identified in Question 1. Foundation model scope: only zero-shot, target-only Chronos-small was evaluated; a larger Chronos variant, a covariate-aware foundation model, or lightweight fine-tuning on this household's own history may close the accuracy gap to the feature-based model while retaining low deployment effort. Single-household evaluation: results reflect one household's usage pattern over one period; generalisation to other households or seasons was not tested. AIC search scope: the SARIMAX grid search was run on a 60-day subsample rather than the full training set for computational tractability, and the seasonal order was fixed rather than searched; a fuller search may find a marginally better-fitting model.

12. Conclusion

This case study compared five classes of forecasting model on a 24-hour-ahead household appliance energy forecasting task. All three more complex approaches — SARIMAX, a feature-based gradient boosting model, and a pretrained foundation model — clearly outperformed simple statistical benchmarks, confirming that the additional modelling effort was justified for this dataset. The feature-based model achieved the best overall accuracy, benefiting from an explicit combination of lag, rolling-window, time-of-day and sensor/weather features. SARIMAX and the Chronos foundation model performed similarly to one another, each with distinct trade-offs around interpretability, uncertainty quantification, and deployment cost. A key caveat throughout is that models using weather covariates were evaluated conditionally on realised rather than forecast weather, which future work should address to obtain a fully realistic operational assessment.

References

- Ansari, A.F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S.S., Pineda Arango, S., Kapoor, S., Zschiegner, J., Maddix, D.C., Wang, H., Mahoney, M.W., Torkkola, K., Gordon Wilson, A., Bohlke-Schneider, M. and Wang, Y. (2024). Chronos: Learning the Language of Time Series. arXiv preprint arXiv:2403.07815.
- Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and Ljung, G.M. (2015). Time Series Analysis: Forecasting and Control. 5th ed. Hoboken: Wiley.
- Candanedo, L.M., Feldheim, V. and Deramaix, D. (2017). Data driven prediction models of energy use of appliances in a low-energy house. *Energy and Buildings*, 140, pp.81–97.
- Chen, T. and Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp.785–794.
- Hyndman, R.J. and Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*. 3rd ed. Melbourne: OTexts.
- Hyndman, R.J. and Koehler, A.B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), pp.679–688.
- Kwiatkowski, D., Phillips, P.C.B., Schmidt, P. and Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1–3), pp.159–178.